Aspect-Based Sentiment Analysis of IMDb Movie Reviews with Evidence-Grounded Silver Labels from a Local LLM

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***Abstract*—Document-level sentiment classification of IMDb reviews is close to saturation, but a single positive or negative label does not say *what* a reviewer liked or disliked. We study aspect-category sentiment analysis on IMDb: for every review, a model predicts one of four labels (positive, negative, neutral, or not mentioned) for six aspects: acting, story, visual, sound, direction and overall. No large aspect-labeled IMDb corpus exists, so we label all 49,777 reviews of a leakage-controlled, movie-grouped split with Qwen3-8B running locally on an 8 GB laptop GPU, at no API cost. The model must first copy a verbatim evidence phrase for each aspect and only then choose a label, and labels whose evidence cannot be found in the review are reset to "not mentioned." On a 300-review reference set this evidence-first prompt raises mean Cohen's κ from 0.54 to 0.72 over a label-only prompt. We then train three multi-head classifiers on the silver labels with head–tail inputs of 512 tokens. Fine-tuned RoBERTa-large reaches a test Macro-F1 of 74.19%±0.13 against the silver labels, ahead of fine-tuned BERT-base (67.86%) and a frozen BERT with a BiLSTM head (62.58%); a movie-clustered paired bootstrap with Holm correction finds every difference significant, overall and for each aspect. Independently of the LLM, its overall predictions agree with the reviewers' own star ratings on 97.5% of decided reviews. Errors concentrate on the neutral class and on reviews longer than 512 tokens. All scores are measured against the LLM labels; a protocol and tooling for human verification of 400 test reviews are released, but the verification itself was not carried out within this project.**

***Index Terms*—aspect-based sentiment analysis, large language models, silver labels, knowledge distillation, RoBERTa, IMDb**

I. Introduction

Movie reviews are a standard testbed for sentiment analysis, and the IMDb Large Movie Review Dataset [1] is its most widely used benchmark. In earlier work on the same data we showed that a fine-tuned RoBERTa-large [2] with head–tail inputs [3] reaches 96.74% Macro-F1 on the official test set for the binary task. At that level, the binary label carries little additional information: a reader who wants to know whether a film is worth watching usually also wants to know *why*. A two-star review may still praise the acting, and a nine-star review may complain about the soundtrack.

Aspect-based sentiment analysis (ABSA) [4], [5] addresses this by assigning sentiment to individual aspects of the reviewed item. For movies, the relevant aspects are well known (acting, story, visual style, music, direction) [6], but there is no large, aspect-labeled corpus of IMDb reviews with a fixed split. Manual annotation of tens of thousands of long reviews is beyond the budget of a student project, and paid LLM APIs were not an option either.

This paper makes four contributions:

- We define a six-aspect, four-label ACSA task on IMDb and label all 49,777 reviews of a movie-grouped split with an open 8B-parameter LLM running locally, using constrained JSON decoding.
- We show that asking the LLM to copy a verbatim evidence phrase *before* each label, and discarding labels whose evidence cannot be found in the review, substantially improves agreement with a reference set (κ 0.54→0.72).
- We train three encoder-based models on the silver labels and compare them with a movie-clustered paired bootstrap; fine-tuned RoBERTa-large is best on every aspect.
- We analyze the errors per aspect, class and review length, use the reviewers' star ratings as an LLM-independent check, and release a protocol and tooling for human verification of the test labels.

II. Related Work

Aspect-based sentiment analysis. SemEval-2014 Task 4 [4] established two ABSA settings: aspect-term sentiment, where the opinion target is a span of text, and aspect-category sentiment (ACSA), where it is one of a fixed set of categories. Our task is ACSA with an explicit "not mentioned" label, so that detection and polarity are learned jointly. Zhang *et al.* [5] survey the many ABSA variants. Thet *et al.* [6] analyzed aspects of movie reviews with lexicon-based methods at the clause level. Sun *et al.* [7] recast ACSA as sentence-pair classification with BERT [8]; we instead use one encoder pass with one classification head per aspect, which is six times cheaper at inference.

LLMs as annotators. Large language models can replace or support human annotators for many text-classification tasks. Wang *et al.* [9] showed that GPT-3 labels can reduce annotation cost substantially, and Gilardi *et al.* [10] found ChatGPT

TABLE I
ASPECTS AND THEIR SCOPE

Aspect	Covers
Acting	performances, cast, casting, voice acting
Story	plot, script, dialogue, characters as written, twists, ending
Visual	cinematography, lighting, special effects, sets, costumes
Sound	music, score, songs, sound effects, audio quality
Direction	the director's work; a name alone is not enough
Overall	the reviewer's verdict on the film as a whole

more accurate than crowd workers on several annotation tasks. Training a smaller model on the outputs of a larger one is a form of knowledge distillation [11]. Unlike these studies, we use a small open model (Qwen3-8B [12]) quantized to 4 bits and served with llama.cpp [13] on a consumer GPU, and we add a verifiable evidence step to reduce hallucinated labels.

Existing movie-aspect data. The Lowerated IMDb review collection [14] provides continuous scores for seven movie aspects on the same IMDb reviews. It has no sound aspect, uses a single value (zero) for both "neutral" and "not mentioned," and does not document how the scores were produced, so we do not use it as ground truth.

III. Task and Dataset

A. Task definition

Each review receives one label from {positive, negative, neutral, none} for each of six aspects (Table I). *Neutral* means that the review discusses the aspect without a clear opinion, or praises and criticizes it about equally; *none* means that the aspect is not discussed. A seventh aspect, pacing, was annotated with the others but is excluded from all experiments.

B. Reviews and split

We use the 50,000 labeled IMDb reviews [1] with the leakage-controlled split of our earlier work: the official 25,000-review test set (3,581 movies) is kept intact, 223 duplicate or leaking training reviews are removed, and the remaining training reviews are split by movie into 21,027 training reviews (2,910 movies) and 3,750 validation reviews (514 movies). No movie occurs in two splits. Each review also keeps its star rating (1–4 or 7–10), which the LLM never sees.

C. Silver labeling with a local LLM

Model and decoding. We label every review with Qwen3-8B [12] quantized to 4 bits (Q4_K_M, 5.0 GB) and served by llama.cpp [13] (build b12081) on the same 8 GB laptop GPU used for training. Decoding is greedy (temperature 0) with the model's reasoning mode disabled, and the output is constrained by a JSON schema, so that every response contains all aspects and only valid labels. Reviews longer than 12,000 characters (three reviews) are shortened to their beginning and end.

Evidence-first prompt. The prompt defines the labels and aspects, lists rules for hard cases (generic praise counts only for *overall*; a named actor without judgment is *neutral* for acting; sarcasm counts by its intended meaning; an explicit score decides *overall*) and gives three short examples. For each aspect the model must first copy the shortest exact phrase (at most 20 words) in which the reviewer talks about that aspect, and only then output the label; with no such phrase, the label must be *none*. After decoding, a non-*none* label is reset to *none* if its evidence is empty or not grounded in the review, i.e., neither a substring of the review nor covered to at least 70% by the review's word tokens. *Overall* is exempt, because a verdict can be implicit. Fig. 1 summarizes the pipeline.

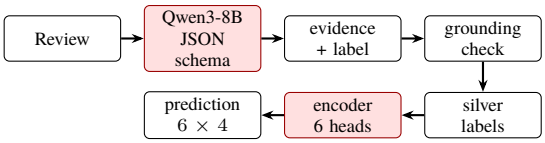


Fig. 1. Labeling and training pipeline. An LLM writes an evidence phrase and a label per aspect; labels with ungrounded evidence are reset to *none*; the resulting silver labels train an encoder with one four-way head per aspect.

TABLE II
AGREEMENT OF QWEN3-8B WITH THE 300-REVIEW REFERENCE SET (COHEN'S κ)

Prompt	Act.	Story	Vis.	Sound	Dir.	Over.	Mean
v2: label only	0.43	0.57	0.57	0.39	0.48	0.77	0.54
v3: evidence first	0.67	0.60	0.70	0.78	0.79	0.77	**0.72**

Prompt development. To choose the prompt we used a reference set of 300 validation reviews (150 positive, 150 negative, random seed 2026). These reviews were labeled by a larger proprietary LLM (Claude) following the written guideline; they are therefore a stronger-model reference rather than human gold labels, and they were used only for prompt development, never for model selection or evaluation. The reference *overall* labels agree with the IMDb star ratings on 99.3% of the 95% of reviews with a decided verdict. Table II compares a label-only prompt (v2) with the evidence-first prompt (v3). The label-only prompt often assigned *neutral* or *negative* to aspects the review never mentions, inferring them from the overall tone; the evidence step removes most of these errors, and mean κ rises from 0.54 to 0.72. Only story remains below 0.6.

Cost and output. With two parallel requests, labeling ran at 0.65 reviews per second (about 21 hours of GPU time for the corpus), with no failed request. The grounding check reset 275 of the roughly 300,000 non-overall labels (0.09%), most of them for sound (105) and story (97). Every record stores the model, prompt version and hash, the evidence phrases and the reset aspects, so that the team can verify labels quickly.

Label distribution. Table III shows the training distribution; validation and test differ from it by less than three points for every cell. Story is criticized 2.3 times more often than praised. Sound, direction and visual are not mentioned in most reviews, and neutral is rare for every aspect, which makes Macro-F1 sensitive to a few hundred neutral examples.

TABLE III
SILVER LABEL DISTRIBUTION ON THE TRAINING SET (%)

Aspect	Positive	Negative	Neutral	None
Acting	36.7	29.2	9.1	25.0
Story	21.4	49.5	15.7	13.4
Visual	19.7	12.1	5.1	63.1
Sound	8.6	6.3	2.1	83.0
Direction	9.3	11.8	3.4	75.4
Overall	43.9	45.7	8.7	1.7

IV. MODELS

All models share one architecture: an encoder reads the review, the representation of the first token passes through dropout and a linear layer with 6×4 outputs, and the output is reshaped into one four-way softmax per aspect. The loss is the sum over aspects of the class-weighted cross-entropy,

$$\mathcal{L} = \sum_{a=1}^{6} w_{y_a}^{(a)} \big(-\log p_a(y_a \mid x)\big), \tag{1}$$

$$w_c^{(a)} = \mathrm{clip}\Big(\sqrt{\tfrac{N}{4\, n_c^{(a)}}},\, 0.2,\, 5\Big), \tag{2}$$

where $n_c^{(a)}$ is the number of training reviews with label c for aspect a. The square-root weights stop the frequent *none* class from dominating visual, sound and direction without overweighting the rarest classes.

Reviews longer than 512 tokens keep the first 128 and the last 382 tokens [3]. This affects 12.3% of test reviews with the BERT tokenizer and 11.8% with the RoBERTa tokenizer. We compare three encoders:

- *Frozen BERT + BiLSTM:* a frozen BERT-base encoder whose token states feed a BiLSTM [15], [16] with 768 units per direction (dropout 0.5); the concatenated final states replace the first-token representation. This mirrors the model of our earlier binary study.
- *BERT-base FT:* BERT-base (uncased, 110M parameters) [8], fully fine-tuned.
- *RoBERTa-large FT:* RoBERTa-large (355M parameters) [2], fully fine-tuned with gradient checkpointing [17] to fit in 8 GB.

All use AdamW [18] with weight decay 0.01, linear warm-up over 10% of the steps followed by linear decay, gradient clipping at 1.0 and BF16 autocast (Table IV). After each epoch we compute the mean Macro-F1 over the six aspects on the validation set and keep the best epoch; the test set is evaluated once per run. Every configuration is trained with seeds 42, 43 and 44 on a laptop RTX 5070 GPU (8 GB) with PyTorch [19] and Transformers [20].

V. EVALUATION PROTOCOL

Metrics. For each aspect we compute Macro-F1 over the four labels (over the labels present in the gold or predicted labels), and the primary score is the mean over the six aspects. Two secondary metrics separate the sub-problems: *detection F1* treats every non-*none* label as "mentioned," and *sentiment Macro-F1* is computed over positive, negative and neutral on the reviews where the silver label is not *none*. Because no human-labeled test set is available, all these scores are measured *against the silver labels*, so they show how well each model reproduces the LLM annotator, not accuracy against human judgment.

An LLM-independent check. The IMDb star rating is written by the reviewer and was never shown to the LLM. For the *overall* aspect we therefore also report the fraction of reviews predicted positive or negative ("decided") and the accuracy of those predictions against the rating (positive for 7–10 stars).

Statistical testing. Reviews of the same movie are not independent, so we use a paired bootstrap [21] that resamples the 3,581 test movies with replacement ($B = 10{,}000$). In each resample we recompute the Macro-F1 of both systems from per-movie confusion counts, average over the three matched seeds and take the difference. We report percentile 95% confidence intervals and two-sided p-values with add-one correction (floor 2×10^{-4}), adjusted with the Holm procedure [22] separately for the three comparisons of the mean score and for the 18 per-aspect comparisons.

TABLE IV
TRAINING SETTINGS (EFFECTIVE BATCH SIZE 32 FOR ALL)

	BiLSTM	BERT-base	RoBERTa-large
Encoder	frozen	fine-tuned	fine-tuned
Learning rate	3×10^{-4}	2×10^{-5}	1×10^{-5}
Micro-batch $\times$ accum.	16×2	8×4	4×8
Epochs	8	3	3
Peak GPU memory	0.9 GB	3.6 GB	6.3 GB
Training time per seed	23 min	15 min	63 min

TABLE V
TEST RESULTS AGAINST SILVER LABELS (MEAN $\pm$ SD OVER THREE SEEDS; MACRO-F1 IN %)

Model	Macro-F1	Detection	Sentiment	Stars
Frozen BERT + BiLSTM	62.58 $\pm$ 0.36	87.71	62.26	94.6
BERT-base FT	67.86 $\pm$ 0.11	90.82	67.09	95.2
RoBERTa-large FT	**74.19 $\pm$ 0.13**	**92.09**	**75.01**	**97.5**

Stars: accuracy of decided *overall* predictions against the IMDb star rating.

VI. RESULTS

A. Main comparison

Table V lists the test results. Fine-tuned RoBERTa-large reaches 74.19% mean Macro-F1, 6.33 points above fine-tuned BERT-base and 11.61 points above the frozen BiLSTM. The ranking holds for detection and for sentiment, but the gap is much larger for sentiment (75.01% vs. 62.26%) than for detection (92.09% vs. 87.71%): finding whether an aspect is discussed is comparatively easy, while judging the opinion requires reading it in context. Run-to-run variation is small; the largest standard deviation over seeds is 0.36 points, for the frozen BiLSTM. On the star-rating check, RoBERTa-large decides 88.8% of the reviews and agrees with the star rating on 97.5% of them.

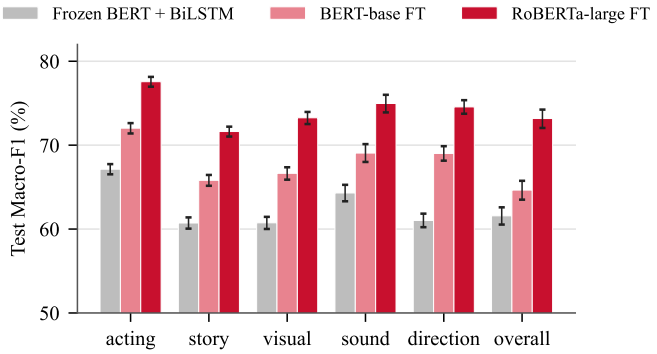


Fig. 2. Test Macro-F1 per aspect against silver labels (mean over three seeds). Error bars are 95% movie-clustered bootstrap intervals.

TABLE VI
PAIRED COMPARISONS OF MEAN TEST MACRO-F1 (POINTS)

Comparison	Δ	95% CI
RoBERTa-large FT vs BERT-base FT	+6.33	[5.99, 6.66]
RoBERTa-large FT vs frozen BiLSTM	+11.61	[11.24, 11.97]
BERT-base FT vs frozen BiLSTM	+5.28	[4.97, 5.59]

Holm-adjusted bootstrap $p = 0.0006$ (the floor for three tests) for all rows.

B. Per aspect and significance

Fig. 2 breaks the score down by aspect. RoBERTa-large is best on all six aspects, from 71.6% on story to 77.6% on acting. Story is the hardest aspect for every model: reviews mix plot summary with judgment, and story has the largest share of neutral labels. Table VI shows that all three pairwise differences of the mean score are significant, with confidence intervals far from zero. The same holds for each of the 18 per-aspect comparisons (Holm-adjusted $p \leq 0.0036$, the floor for 18 tests); the smallest is BERT-base over the frozen BiLSTM on *overall*, +3.07 points [2.21, 3.91].

C. Where the errors are

Neutral is the weak class. Fig. 3 shows per-class F1 for RoBERTa-large. Positive and negative reach 75–94%, and *none* exceeds 92% for the three rarely mentioned aspects, but neutral stays between 35.5% (sound) and 53.5% (overall). The larger encoder helps most exactly here: for neutral, RoBERTa-large improves on BERT-base by 9–17 points per aspect, for example from 23.5% to 38.8% on visual. The *none* class of *overall* is also weak (51.5%), but it covers only 1.7% of the reviews.

Detection versus polarity. The error types differ by aspect. For visual, sound and direction, most RoBERTa-large errors are about whether the aspect is mentioned at all: 66–73% of the errors involve *none* on one side (*none*→negative alone accounts for 17–18%). For acting and story, most errors involve neutral (55% and 64%; silver neutral predicted positive alone is 16% of the story errors), and for *overall* 78% do. Confusions between positive and negative are rare: 2–16% of the errors depending on the aspect.

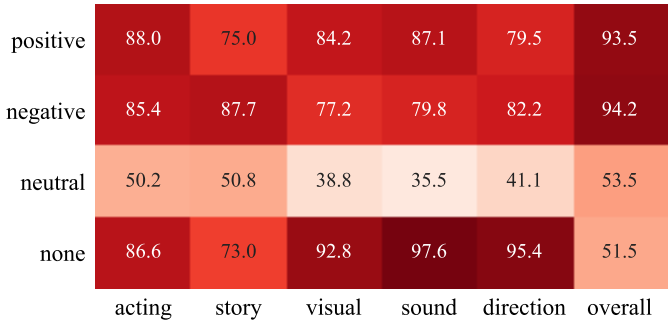


Fig. 3. Per-class test F1 (%) of RoBERTa-large for each aspect, mean over three seeds.

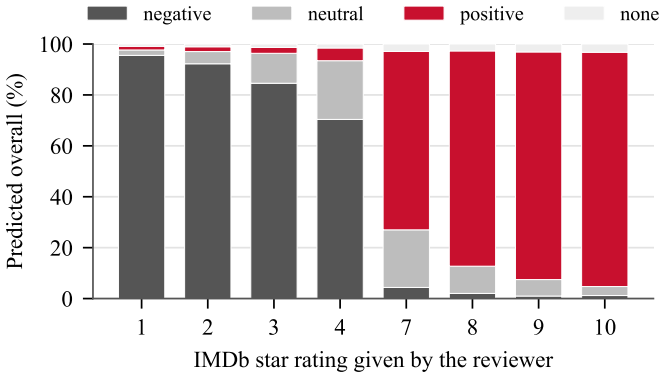


Fig. 4. Distribution of RoBERTa-large's *overall* prediction by the reviewer's IMDb star rating (test set, three seeds pooled).

Long reviews. Reviews longer than 512 tokens are clearly harder for all models: RoBERTa-large scores 75.1% on reviews that fit and 66.8% on those that do not; the gap is 7.4 points for BERT-base and 5.8 points for the BiLSTM. In our binary study, the same head–tail input closed this gap completely for RoBERTa-large. For aspects, an opinion about the music or the camera work can sit in the discarded middle, so a single verdict at the end of a review no longer suffices.

Stars as a sanity check. Fig. 4 plots RoBERTa-large's *overall* prediction against the star rating. Extreme ratings are almost always predicted with the matching polarity (95.5% negative at one star, 92.0% positive at ten stars). Neutral predictions concentrate at the boundary ratings: 23.1% of four-star and 22.6% of seven-star reviews, compared with 2.1% of one-star reviews. This pattern was not supervised by the ratings and matches what one would expect from mixed reviews, which suggests that the neutral class is meaningful even though it is hard to predict.

Consistent disagreements. For each aspect we counted the test reviews on which all nine runs (three models, three seeds) predict the same label and that label differs from the silver one. These reviews make up 1.6% (sound) to 6.5% (story) of the test set, and the most frequent pattern for story is a silver *neutral* that all models read as negative (369 reviews) or positive (336). Such cases are candidates for silver-label errors as well as for systematic model errors. A short inspection finds both. In one

one-star review the LLM labeled *overall* as neutral, quoting "It was okay," although the reviewer writes this only as what they would have thought before the ending, which then explains the review's "undeniably hostile tone"; all models predict negative, which agrees with the rating. In another, the LLM correctly labeled sound negative from the phrase "a bad music," which all nine runs missed. These reviews should be prioritized in any human verification.

VII. Human Verification Protocol

Because the silver labels come from an LLM, scores against them measure agreement with that annotator, not accuracy against human judgment. We prepared, but did not carry out within the project timeline, a verification protocol for three annotators. It samples 400 test reviews stratified by star rating, with at most one review per movie (mean length 232 words). Each annotator labels 200 reviews: 100 shared by all three, used to measure inter-annotator agreement with Fleiss' κ [23], and 100 of their own. For shared reviews the human label is the majority vote; reviews without a majority are excluded for that aspect. The released tooling creates the annotation workbooks and computes inter-annotator κ, the agreement of the Qwen3-8B labels with the human labels, and the Macro-F1 of every trained model against them. By default annotators see the LLM label and evidence, which speeds up labeling but may anchor them toward the LLM; a blind mode hides both and gives an unbiased estimate of LLM accuracy.

Until such a verification is done, the only evidence independent of the LLM is the star-rating check of Section VI, which covers the *overall* aspect only.

VIII. Discussion

Cost. The whole pipeline ran on one laptop: about 21 hours of GPU time for labeling and 5 hours for the nine training runs. Running the labeler locally was what made the project feasible, and constrained decoding removed the need for any output repair: no request failed.

Evidence as a cheap control. The evidence step helped in two ways. It made the LLM commit to a specific phrase, which removed most labels inferred from overall tone, and it gave annotators a direct pointer for verification. The automatic grounding check reset few labels (0.09%); most of the gain came from the instruction itself rather than from the filter.

Limitations. (i) All main results are measured against silver labels, and the models are bounded by the quality of those labels; the reference set used for prompt development was itself labeled by an LLM, and no human verification of the test labels was carried out, so we cannot report accuracy against human judgment for any aspect except the star-rating check on *overall*. (ii) Agreement on story (κ=0.60) is the lowest, so story results are the least reliable. (iii) The neutral class is rare and its boundary is subjective. (iv) Reviews longer than 512 tokens lose the middle part. (v) We study one corpus in one language.

Future work. The first step is to carry out the human verification protocol of Section VII, preferably in blind mode. Further steps are encoders that read whole reviews [24], aspect-specific attention that also returns evidence, filtering or re-weighting silver labels by the agreement between the LLM and the trained models, and fine-tuning on a human-verified subset.

IX. Conclusion

We built a six-aspect sentiment dataset over all 49,777 reviews of a movie-grouped IMDb split by labeling it with a local 8B-parameter LLM, and showed that an evidence-first prompt with a grounding check improves agreement with a reference set from κ=0.54 to 0.72. Trained on these silver labels, a fine-tuned RoBERTa-large reaches 74.19% mean Macro-F1, significantly ahead of BERT-base and a frozen BERT with a BiLSTM on every aspect, and its *overall* predictions agree with the reviewers' star ratings on 97.5% of decided reviews. The remaining errors are concentrated on the neutral class and on long reviews.

Acknowledgment

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